PRICING A EUROPEAN CALL OPTION

Learning Outcome

- explain how to value a derivative using a one-period binomial model

Consider a one-year European call option with an exercise price (X) of €100. The underlying spot price (S_0) is €80. The one-period binomial model corresponds to the option's time to expiration of one year.

The binomial model specifies the possible values of the underlying asset in one year, where the option value is a known function of the value of the underlying asset. Further, we assume that $S_1^d < X < S_1^u$ (i.e., the exercise price of the option [X = €100] is between the value of the underlying in the two scenarios)—for example, setting S_1^d = €60 and S_1^u = €110. Then, $R^u = 1.375$ and $R^d = 0.75$. The value of the call option is as follows:

- At $t = 0$, the call option value is c_0.
- At $t = 1$, the call option value is either c_1^u (if the underlying price rises to S_1^u) or c_1^d (if the underlying price falls to S_1^d).
  - **Up move to S_1^u:** Call option ends up *in-the-money*

    $$c_1^u = \text{Max}(0,\ S_1^u - X) = \text{Max}(0,\ €110 - €100) = €10.$$

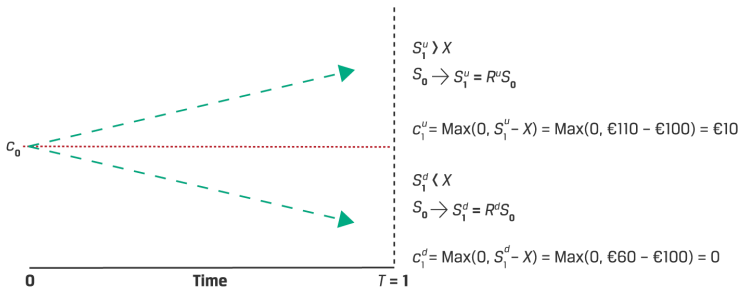
  - **Down move to S_1^d:** Call option expires *out-of-the-money*

    $$c_1^d = \text{Max}(0,\ S_1^d - X) = \text{Max}(0,\ €60 - €100) = 0.$$

Exhibit 2 summarizes the one-period binomial model and the value of the underlying asset and call option.

Exhibit 2:

One-Period Binomial Option Pricing



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The only unknown value is c_0, which may be determined using replication and no-arbitrage pricing. That is, the value of both the option and its underlying asset in each future scenario may be used to construct a risk-free portfolio. For example, assume that at $t = 0$ we sell the call option at a price of c_0 and purchase h units of the underlying asset. Denoting the value of the portfolio as V, we have the following:

$$V_0 = hS_0 - c_0 \quad (3)$$

$$V_1^u = hS_1^u - c_1^u = h \times R^u \times S_0 - \text{Max}(0,\ S_1^u - X) \quad (4)$$

$$V_1^d = hS_1^d - c_1^d = h \times R^d \times S_0 - \text{Max}(0,\ S_1^d - X) \quad (5)$$

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Valuing a Derivative Using a One-Period Binomial Model

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so that $V_1^u = V_1^d$ (i.e., the portfolio value is the same in either scenario). The value of the combination, V_1, will not change if the underlying asset price changes. For portfolio V_0, the impacts of the changes are as follows:

- **Up move from S_0 to S_1^u by R^u:**
  - The asset value changes from €80 to €110 by 1.375, or 37.5%.
  - The call option ends up *in-the-money*:

$$c_1^u = \text{Max}\left(0,\ S_1^u - X\right) = \text{Max}(0, €110 - €100) = €10.$$

  - Total portfolio value: $V_1^u = hS_1^u - c_1^u$ = €110 × h − €10.
- **Down move from S_0 to S_1^d by R^d:**
  - The asset value changes from €80 to €60 by 0.75, or –25%.
  - The call option expires *out-of-the-money*:

$$c_1^d = \text{Max}\left(0,\ S_1^d - X\right) = \text{Max}(0, €60 - €100) = 0.$$

  - Total portfolio value: $V_1^d = hS_1^d - c_1^d$ = €60 × h.

Since we create two portfolios at time t = 0 with identical payoffs at option expiry at time t = 1, we must solve for h, the ratio between the underlying asset, S_0, and the call option, c_0, such that $V_1^u = V_1^d$, or $hS_1^u - c_1^u = hS_1^d - c_1^d$. Solving for h^* yields:

$$h^* = \frac{c_1^u - c_1^d}{s_1^u - s_1^d}, \qquad 6$$

where all quantities on the right-hand side are known at t = 0. Equation 6 gives us the **hedge ratio** of the option, or the *proportion* of the underlying that will offset the risk associated with an option. In our sold call option example,

$$h^* = \frac{c_1^u - c_1^d}{s_1^u - s_1^d} = \frac{€10 - €0}{€110 - €60} = \frac{10}{50} = 0.20.$$

For each call option unit sold, we buy 0.2 units of the underlying asset (or for each underlying asset unit, we must sell 5 call options to equate the portfolio values at t = 1). Consider the two scenarios as follows:

- **Up move from S_0 to S_1^u by R^u:**
  - Total portfolio value: $V_1^u = hS_1^u - c_1^u$

    = €110 × 0.20 − €10 = €22 – 10 = €12.
- **Down move from S_0 to S_1^d by R^d:**
  - Total portfolio value $V_1^d = hS_1^d - c_1^d$ = €60 × 0.20 = €12.

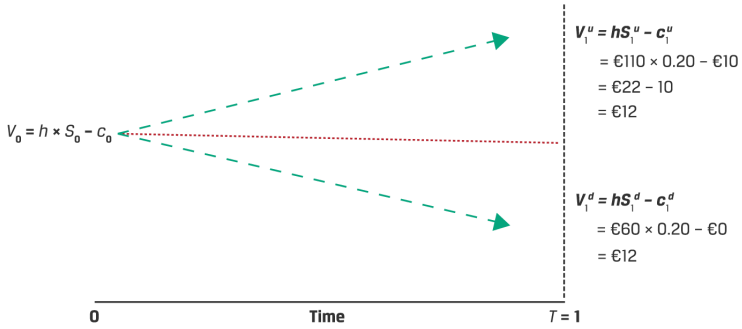
The portfolio values are the same, $V_1^u = V_1^d$, which has two implications:

1. We can use either portfolio to value the derivative, and
2. The return $V_1^u / V_0 = V_1^d / V_0$ must equal one plus the risk-free rate.

Exhibit 3 summarizes these results.

Exhibit 3:

Value of the Hedged Portfolio



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This hedging approach is not specific to call options and can be used for any derivative contract whose value is entirely determined by the underlying asset's value at $t = 1$.

To prevent arbitrage, the portfolio value at $t = 1$,

$$V_1 = h \times R^u S_0 - c_1^u = h \times R^d S_0 - c_1^d,$$

should be discounted at the risk-free rate so that:

$$hS_0 - c_0 = \frac{V_1}{1+r}.$$

Based on no-arbitrage pricing and the certain portfolio payoff, V_1, the value of the call option may be shown as:

$$c_0 = h \times S_0 - V_1(1+r)^{-1}. \qquad 7$$

Substituting the information from the previous example, we already established that V_1 = €12. For an annual risk-free rate of 5%:

$c_0 = h \times S_0 - V_1(1 + r)^{-1}$

$=0.2 \times €80 - €12\ (1.05)^{-1} = €16 - €11.43 = €4.57.$

The call option price c_0 is therefore €4.57. We may confirm an investor in the hedge portfolio would earn the risk-free rate (r) of 5% by comparing the initial portfolio value V_0 with the portfolio value after one period of V_1 (recalling that $V_1^u = V_1^d$) equal to €12:

$V_0 = h \times S_0 - c_0$

V_0 = €11.43 = €16 - €4.57

$V_1 = V_0(1 + r)$ 8

V_1 = €12 = €11.43 (1 + 0.5)

EXAMPLE 1

Hightest Capital believes that a particular non-dividend-paying stock is currently trading at $50 and is considering the sale of a one-year European call option at an exercise price of $55. Answer the following questions:

1. If the stock price is expected to either go up or down by 20% over the next year, what price should Hightest expect to receive for the sold call option? Assume a risk-free rate of 5%.

   Hide Solution

   Solution:

   Denote the initial price of the underlying stock as S_0= $50 and the exercise price of the call option as X = $55. If the stock price moves up or down by 20%, then:

   $$S_1^u = R^u S_0 = 1.2 \times \$50 = \$60.$$

call option at expiry is Max(0, $S_T - X$):

$$c_1^u = \text{Max}(0,\ S_1^u - X) = \text{Max}(0,\ \$60 - \$55) = \$5.$$

$$c_1^d = \text{Max}(0,\ S_1^d - X) = \text{Max}(0,\ \$40 - \$55) = \$0.$$

The call option value is $5 if the underlying stock increases in price by 20% and zero if it decreases by 20%. The hedge ratio of the option is:

$$h^* = \frac{(c_1^u - c_1^d)}{(S_1^u - S_1^d)} = \frac{\$5.00 - \$0}{\$60.00 - \$40.00} = \frac{\$5.00}{\$20.00} = 0.25.$$

To create a risk-free portfolio, we can sell a call option and purchase 0.25 units of the underlying asset today. At maturity, the hedged portfolio value is:

$$V_1 = V_1^u = h^* S_1^u - c_1^u = 0.25 \times \$60.00 - \$5.00 = \$10.00.$$

$$V_1 = V_1^d = h^* S_1^d - c_1^d = 0.25 \times \$40.00 - \$0.00 = \$10.00.$$

We can use either V_1^u or V_1^d to compute the certain value, V_1, and the present value of the hedged position today is:

$$V_0 = V_1(1 + r)^{-1} = \$10.00 \times 1.05^{-1} = \$9.52.$$

The call option value is:

$$c_0 = h^* S_0 - V_0 = 0.25 \times \$50.00 - \$9.52 = \$12.50 - \$9.52 = \$2.98.$$

The no-arbitrage price of the call option should be $2.98. If not, then investors would be able to construct a synthetic risk-free asset using the option and its underlying asset with a higher return than the risk-free rate.

Note that the hedge ratio (0.25 units in this example) is positive in this case since the derivative is sold and the underlying is purchased. A negative hedge ratio implies that *both* the derivative and the underlying are purchased or sold to create the hedge. Also, while the hedge ratio is usually expressed as a fraction of an underlying unit, option contracts are usually traded in larger size, allowing a round number of underlying assets to be purchased or sold as a hedge. It is important that the *ratio* (4:1 in this case) of options to underlying units is maintained in the portfolio.

2. How would the call option price change if the stock price were expected to either go up or down by 40% over the next year?

Hide Solution

Solution:

If the stock price changes by 40%, then the call option payoff at expiry is:

$$c_1^u = \text{Max}(0,\ S_1^u - X) = \text{Max}(0,\ \$70 - \$55) = \$15.$$

$$c_1^d = \text{Max}(0,\ S_1^d - X) = \text{Max}(0,\ \$30 - \$55) = 0.$$

The hedge ratio of the option is:

$$h^* = \frac{(c_1^u - c_1^d)}{(S_1^u - S_1^d)} = \frac{\$15 - \$0}{\$70 - \$30} = \frac{\$15}{\$40} = 0.375.$$

At maturity, the perfectly hedged portfolio is worth:

$$V_1 = V_1^u = h^* S_1^u - c_1^u = 0.375 \times \$70 - \$15 = \$11.25.$$

$$V_1 = V_1^d = h^* S_1^d - c_1^d = 0.375 \times \$30 - \$0 = \$11.25.$$

The present value of the hedged position at $t = 0$ is:

$$V_0 = V_1(1 + r)^{-1} = \$11.25 \times 1.05^{-1} = \$10.71.$$

Then, the call option price at $t = 0$ is:

range of future price changes increases the value of the call option.

3. If Hightest had a more optimistic outlook on the future stock price (i.e., they estimated a higher probability of the option ending up in-the-money), how would the expected call option price change?

Hide Solution

Solution:

Since the actual probabilities of an up or a down move in the underlying asset do not affect the no-arbitrage value of the option, the option price that Hightest may charge should not change. Hightest can offset the risk of selling the call by purchasing h^* units of the underlying asset, so any directional views on the stock price do not affect the hedge position.

4. What would be the price of a one-year put option at an exercise price of $55 if the stock price were expected to change by 20%? 40%?

Hide Solution

Solution:

Using put–call parity, $c_0 - p_0 = S_0 - X(1 + r)^{-T}$, and rearranging terms to solve for the put price, when the stock price changes by 20% and the call option price is $2.98, the result can be calculated as:

$$p_0 = c_0 - S_0 + X(1 + r)^{-T} = \$2.98 - \$50 + \$55(1 + 0.05)^{-1} = \$5.36.$$

When the underlying stock price changes by 40% and the call option price is $8.04, the put option price is:

$$p_0 = c_0 - S_0 + X(1 + r)^{-T} = \$8.04 - \$50 + \$55(1 + 0.05)^{-1} = \$10.42.$$

Question Set: Binomial Valuation of Options

1. Determine the correct answers to fill in the blanks: To price a contingent claim, such as an option, a model for the _____ of the underlying asset is needed due to the _____ nature of the contract's payoff.

Hide Solution

Solution:

To price a contingent claim, such as an option, a model for the *future price behavior* of the underlying asset is needed due to the *asymmetric* nature of the contract's payoff.

2. Describe the main difference between pricing a contingent claim and pricing a forward commitment.

Hide Solution

Solution:

The symmetric nature of a forward commitment's payoff (i.e., the obligation to transact at maturity) allows the commitment to be perfectly replicated without the need to model the future price behavior of the underlying asset. However, the asymmetric nature of a contingent claim's payoff (i.e., the right but *not* the

3. If a one-period binomial model is used to price an at-the-money put option, which of the following statements is *most* accurate? The option will be:

A. in-the-money if the price moves up.
B. out-of-the-money if the price moves up.
C. at-the-money if the price moves up or down.

Hide Solution

Solution:

The correct answer is B; the option will be out-of-the-money if the price moves up. An at-the-money put option has an exercise price equal to the underlying asset price. Therefore, a price decrease will result in the put option moving *in the money*, and a price increase will result in the put option moving *out of the money*.

4. Explain how increasing the up gross return, R^u, and/or decreasing the down gross return, R^d, in a one-period binomial model would influence the price of a call and a put option.

Hide Solution

Solution:

In a one-period binomial model, the volatility of the underlying asset is represented by the spread between the up gross return, R^u, and the down gross return, R^d. Therefore, if either the up gross return increases or the down gross return decreases (or both), the price of the underlying asset at maturity will be more volatile. If all else remains equal, then the price of both call and put options will increase when the underlying asset is expected to have a higher volatility over the life of the option.

5. A put option on a non-dividend-paying stock has an exercise price, X, of £21 and six months left to maturity. The current stock price, S_0, is £20, and an investor believes that the stock's price in six months' time will be either 10% higher or 10% lower.

a. Describe how the investor can construct a perfectly hedged portfolio using the put option and its underlying stock.
b. What will the value of the hedged portfolio be in the scenario that the stock price rises and the scenario that the stock price falls (assume a risk-free rate is 4%)?
c. What is the no-arbitrage price of the put option?

Hide Solution

Solution:

a. Denote the initial price of the underlying stock as S_0= £20.00 and the exercise price of the put option as X = £21.00. If the stock price moves up by 10%, then:

$$S_1^u = R^u S_0 = 1.1 \times £20.00 = £22.00.$$

If the stock price moves down by 10%, then

$$S_1^d = R^d S_0 = 0.9 \times £20.00 = £18.00.$$

Given that the payoff of a put option at expiry is Max(0, $X - S_T$):

$$p_1^u = \text{Max}\left(0,\ X - S_1^u\right) = \text{Max}(0,\ £21.00 - £22.00) = £0.$$

$$p_1^d = \text{Max}\left(0,\ X - S_1^d\right) = \text{Max}(0,\ £21.00 - £18.00) = £3.00.$$

$$h^* = \frac{(p_1^u - p_1^d)}{(S_1^u - S_1^d)} = \frac{£0 - £3}{£22 - £18} = -\frac{£3}{£4} = -0.75.$$

So, to create a risk-free portfolio, we can buy the put option and buy 0.75 units of the underlying asset.

b. At maturity, the value of the perfectly hedged portfolio is:

$$V_1 = V_1^u = h^* S_1^u + p_1^u = 0.75 \times £22 + £0 = £16.50.$$

$$V_1 = V_1^d = h^* S_1^d + p_1^d = 0.75 \times £18 + £3.00 = £16.50.$$

We can use either V_1^u or V_1^d to compute the certain value, V_1, and the present value of the hedged position today is:

$$V_0 = V_1(1 + r)^{-1} = £16.50 \times 1.04^{-0.5} = £16.18.$$

c. The price of the put option is:

$$p_0 = V_0 - h^* S_0 - = £16.18 - 0.75 \times £20 = £16.18 - £15 = £1.18.$$

The no-arbitrage price of the put option should be £1.18.

6. Determine the correct answers to fill in the blanks: When the _____ applies, the rate of return on all (real or synthetic) risk-free assets should equal the _____.

Hide Solution

Solution:

When the *law of one price* applies, the rate of return on all (real or synthetic) risk-free assets should equal the *risk-free rate*.

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AZ

The price of the put option is:
p0 = V0 − h*S0− = £16.18 − 0.75 × £20 = £16.18 − £15 = £1.18.

Since hedge ratio was negative, shouldn't it be 16.18 -(-0.75×20)?

Created 3 months ago by Abdulaziz Zahid — 0 replies | Last Activity: 3 months ago

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SG

On equation 8 V1 = €12 = €11.43 (1 + 0.5), I believe 0.5 should be 0.05 (5%)

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KR

is the sign for a hedge ratio relevant?

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